

[KTW22,ESORICS] Anonymous Traceback for E2EE

SUMMARY: Based on [TMR19,CCS], the authors implement anonymous path and source traceback with minimal modification.

1 Background

While TMR19 achieves traceability for EEMS, it does not support anonymous messaging system. And, TMR19 reveals the communication parties to the platform which reveals the sociogram of users to the platform.

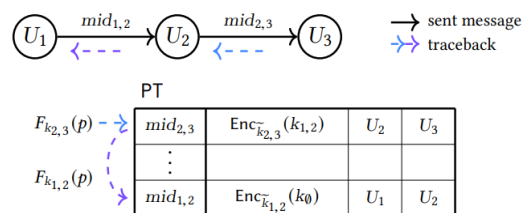
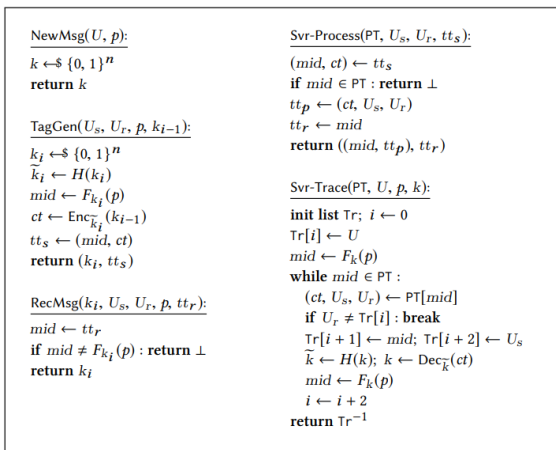
2 Setting & Goals

- Besides the security goals (i.e., confidential and accountability) of TMR19, this work provides anonymity for users in addition.
- This work does not provide deniability, since a signature is contained in message for recipients to authenticate the identity of senders.

3 Methodology | Roadmap

To show the two schemes of this paper, we begin with the path traceback scheme in TMR19.

3.0 Path traceback in TMR19



- $k_{\{i\}}$: one time key (OTK)
- mid: one time message tag from OTK (identify&bind plaintext and sender)
- Enc: link the sender and its previous node

- Random oracle is used to block

3.1 Anonymous path traceback

Based on the above scheme, the idea of this anonymous scheme is straightforward.

1. C_{pk} : Encrypting the identity of the sender with OTK $k_{\{i\}}$ to preserve anonymity;
2. Sig : Signing the whole message with the sender's pk, and encrypting the signature to avoid identity leakage;
3. ts : using timestamp to avoid replace attack.

Send tag	C_k	C_{pk}	C_{sig}	timestamp
$F_{k_i}(p) \rightarrow mid$	$Enc_{k_i}(k_{i-1})$	$Enc_{k_i}(pk_s)$	$Enc_{k_i}(Sig_{sk_s}(mid, C_{pk}, C_k, ts))$	ts

3.2 Anonymous source traceback

The above scheme reveals the path from the reporter to the root to the platform, which undermines E2EE privacy. Therefore, this part tries to reveal only the originator to the platform.

For a trace-based scheme, anonymous source traceback is impossible in the single-server setting (Because the trace algorithm is step-by-step). Naturally, the authors divided the server to non-colluding serve to store trace and message information respectively.

The additional ephemeral signature is provided to the trace server to verify the validity of the query from the message server.

Trace Server	tag	C_k	timestamp	One-time key
	mid	$Enc_{k_i}(k_{i-1})$	ts	$pk_{\{eph\}}$

Message Server	tag	C_{pk}	C_{sig}	timestamp	One-time key	One-time Sig
	mid	$Enc_{k_i}(pk_s)$	$Enc_{k_i}(Sig)$	ts	$pk_{\{eph\}}$	$Sig_{\{eph\}}$

4 Evaluation

The performance of the two schemes is very similar to TMR19, since they do not introduce additional heavy loaded cryptographic operations.

5 Observation & Insights

- The authors argue that they do not design anonymous tree traceback because tree traceback reveals too much (unnecessary) information to the platform. However, from my observation, this probably because their solution cannot achieve anonymous tree traceback trivially.
- This work is straightforward. What the author does is exactly what the reader thinks. There is no specific challenge in this work.